

MARYAM SOHAIL AHMED

Machine Learning · AI Engineering | Karachi, Pakistan
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SUMMARY

BS Artificial Intelligence student (CGPA 3.69) who builds and evaluates machine learning systems end to end and reports results together with their limits. Former Machine Learning intern at FlyRank AI and author of an SSRN preprint on why accuracy alone picks the wrong medical-imaging model. Seeking an ML / AI internship.

EXPERIENCE

Machine Learning Intern, FlyRank AI

Completed Sep 2026

- Built an end-to-end content opportunity scoring pipeline, from problem framing to a deployed research write-up; a Random Forest reached Precision@50 of 0.76 and overtook the Logistic Regression baseline.
- Found and corrected a labelling flaw mid-project, and documented that both models fell below the base rate on the May holdout, pointing to data drift.
- Completed FlyRank's General AI Fluency track (agent design, MCP) and built a Claude-based deadline triage agent.

PROJECTS

Chest X-Ray Pneumonia Detection: CNN Comparison (SSRN preprint)

Sep 2026

Python, TensorFlow, Keras

- Compared three CNNs on the real RSNA dataset (26,684 X-rays). MobileNetV2 transfer learning reached AUC 0.838 and 85% pneumonia recall, while the higher-accuracy Deep CNN (79.1%) caught only 39% of pneumonia cases.
- Caught an impossible earlier result (AUC 1.0 on synthetic data), retrained on real data, and published the findings.

KarachiWatts: Load-Shedding Predictor

Python, scikit-learn, Streamlit, Groq (LLaMA 3.3 70B)

- Built a bilingual (Urdu and English) chatbot that answers outage questions from a Random Forest model's prediction instead of generating its own numbers, and deployed it live on Streamlit.
- Compared three regression models for outage duration across 15 Karachi neighborhoods; Random Forest scored best (R^2 0.991, MAE 0.26 h) on 16,440 synthetic records, which shows the pipeline works, not real-world accuracy.

Drift Watch: MLOps Sentinel (IBM TechXchange 2026 Hackathon, solo)

Aug 2026

Python, Streamlit, IBM Bob

- Prototype that audits ML pipelines for silent failures such as out-of-range inputs and target leakage, and drafts remediation patches.

SkyGuard PK: Flood and Heatwave Alerts (IBM AI Builders Challenge)

Aug 2026

Python, scikit-learn, NASA POWER API, Streamlit

- Urdu and English risk alerts for Pakistani cities from NASA satellite data and a Random Forest classifier, with adjustable thresholds.

Also built: [Al-Bayan](#) (React and Flask streaming tafseer chat), [Samajh AI](#) (bilingual AI-literacy tutor), [Sentiment Analyzer](#) (pretrained DistilBERT with confidence thresholding, pytest), [TeamTasks](#) (Laravel 11 Kanban with role-based access). More on [GitHub](#).

SKILLS

ML and data: Python, scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face, pandas, NumPy, Matplotlib, Tableau

LLM applications: Groq (LLaMA 3.3), Claude API, Model Context Protocol, LangChain

Apps and tools: Streamlit, Flask, React, Laravel, Git, SQL, pytest, Vercel

EDUCATION

BS Artificial Intelligence, Dawood University of Engineering & Technology, Karachi

Oct 2024 – Present

CGPA 3.69. Coursework: machine learning, neural networks, data structures and algorithms, databases, linear algebra and calculus.

Intermediate (Pre-Engineering), Liaquat Govt. Girls College, Karachi

2023

COMPETITIONS AND CERTIFICATIONS

- **Deloitte Data Analytics Job Simulation** (Forage, Jun 2026): analysed factory telemetry in Tableau; classified pay equality in Excel.
- **LangChain Mysteries, PROCOM'26** (FAST NUCES, Feb 2026): team of four; speech-to-text, document forensics and AI reasoning.
- **Top scorer**, Skillo x Sync Backend and AI Engineering Intern Challenge.
- **Certificates:** Anthropic Academy (Claude API, MCP, Agent Skills); IBM Introduction to Artificial Intelligence.